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ORIGINAL ARTICLE

Right and Left Atrial Dysfunction as Independent Cardiovascular Risk Factors: A UK Biobank Study

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BACKGROUND: Atrial cardiopathy often precedes atrial fibrillation (AF) and has emerged as an independent risk factor for cardiovascular outcomes. However, previous studies have been limited in size and have overlooked the right atrium.

METHODS: In 51 693 UK Biobank participants without prevalent AF, we assessed biatrial volumes and emptying fraction from cardiac magnetic resonance imaging using deep learning segmentation. We evaluated associations with new-onset AF, ischemic stroke, heart failure, and dementia, conducted a genome-wide association study, and evaluated causal associations using Mendelian randomization.

RESULTS: Among 51 693 adults, the mean (SD) age was 65 (7.7) years, and 24 584 (48%) were male. During the 4-year follow-up, 964 (1.9%) developed AF, 266 (0.5%) developed ischemic stroke, 365 (0.7%) developed heart failure, and 72 (0.1%) developed dementia. After adjustment for clinical risk factors, both left and right atrial measures were associated with new-onset AF (left atrial minimal volume; hazard ratio, 1.55 [95% CI, 1.48–1.62]), ischemic stroke, and heart failure, with stronger associations in women. Left atrial minimal volume was also associated with dementia. Our genome-wide association study identified 51 (27 novel) genetic associations with atrial measures, many of which do not overlap with established AF loci. Genetic correlations revealed that each atrium had varying correlations with cardiometabolic risk factors, and Mendelian randomization demonstrated that left atrial measures had direct causal effects on AF and stroke risk. However, the stroke associations were attenuated after accounting for AF variants.

CONCLUSIONS: In this largest assessment of biatrial structure and function to date, both left and right atrial cardiopathies were independently associated with increased risk of adverse cardiovascular events. We identified several novel genetic loci for atrial traits and observed unique genetic correlations between left and right atrial traits and cardiovascular phenotypes, providing insight into chamber-specific remodeling. Several of these measures are likely to be causal determinants of cardiovascular complications previously attributed to AF.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: atrial fibrillation ■ cardiovascular diseases ■ deep learning ■ heart atria ■ heart failure

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Atrial fibrillation (AF), the most common cardiac arrhythmia, affects over 59 million people worldwide and is a significant risk factor for heart failure, ischemic stroke, dementia, and mortality.[1–3] Treating AF with anticoagulation can reduce the risk of ischemic stroke by two-thirds.[4] Still, many patients with AF remain untreated, and emerging evidence suggests that some of the morbidity and mortality attributed to AF may result from underlying structural and contractile abnormalities of the cardiac atria, referred to as atrial cardiopathy.[5,6] Recent studies have found that sensitive measures of impaired left atrial function from echocardiography and cardiac magnetic resonance imaging (CMR) are stronger independent risk factors for AF, ischemic stroke, vascular brain injury, and

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WHAT IS KNOWN?

- In the world's largest cardiac imaging study, left and right atrial structure and function measures were independently associated with increased risk of atrial fibrillation (AF), ischemic stroke, heart failure, and dementia, with stronger associations in women.
- Genome-wide association analyses identified 51 loci associated with left and right atrial function measures, several of which are not known atrial function or AF loci.
- Mendelian randomization analyses supported a causal relationship between left atrial measures and AF, ischemic stroke, and cardioembolic stroke; however, we cannot exclude the possibility that these associations are indirect effects mediated through AF.

WHAT THE STUDY ADDS

- Sensitive measures of biatrial cardiopathy from cardiac magnetic resonance imaging are potent risk factors for AF, ischemic stroke, and heart failure, independent of established risk factors for these outcomes. These measures may inform future risk-stratification efforts among individuals at high risk for adverse cardiovascular outcomes, especially women who are at the greatest risk for AF-related complications.
- The identification of distinct genetic loci and varying genetic correlations associated with biatrial measures provides novel insight into the underlying genetic pathways and chamber-specific remodeling and may inform future therapeutic targets.

Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
APOE4	apolipoprotein E4
BMI	body mass index
CMR	cardiac magnetic resonance imaging
CRP	C-reactive protein
GWAS	genome-wide association study
HDL	high-density lipoprotein
LAEF	left atrial emptying fraction
$LAVI_{max}$	left atrial maximal volume
$LAVI_{min}$	left atrial minimal volume
MR	Mendelian randomization
SBP	systolic blood pressure
UKB	UK Biobank

dementia than electrocardiographic measures reflecting left atrial structure.[7–10] In the absence of randomized controlled trials of interventions targeting atrial cardiopathy, it remains unclear whether these relationships are causal.

Most prior research on atrial cardiopathy has focused on the left atrium despite evidence of the right atrium's critical role in AF.[6,7,9,11] As the lowest-pressure cardiac chamber, the right atrium is highly susceptible to hemodynamic changes, potentially allowing for earlier identification of cardiac dysfunction.[12] Recent epidemiological studies have reported that structural changes in the right atrium are associated with an increased risk of AF, even after adjusting for left atrial parameters.[13] Similar findings have been reported in clinical settings, where right atrial measures were superior to left atrial measures in predicting AF recurrence.[14] However, little is known about the potential impact of right atrial structure and function on other cardiovascular diseases, including ischemic stroke and heart failure.

To advance our understanding of the role of both left- and right-sided atrial cardiopathies in the development of clinical complications, we applied deep learning image analyses to CMR performed on nearly 60 000 participants of the UK Biobank (UKB) Imaging Study and evaluated associations with longitudinal clinical outcomes. To explore the biology underlying these potential causal relationships, we used these data to perform the largest genome-wide association study (GWAS) of atrial function to date, evaluated genetic correlations between phenotypes, and assessed for evidence of causal associations with Mendelian randomization (MR).

METHODS

Data that support the findings of this study are available to approved investigators by application to the UKB (www.ukbiobank.ac.uk).

Study Cohort

The UKB is a prospective cohort study that recruited 500 000 participants aged 40 to 69 years from the United Kingdom between 2006 and 2010.[15] The baseline assessment included comprehensive questionnaires, biochemical measurements, anthropometry, and interviews.[15] All UKB participants underwent whole-genome genotyping through the UKB Whole-Genome Sequencing Consortium.[16] All participants provided electronic informed consent at the UKB assessment centers. The UKB received ethics approval from the North West Multi-Center Research Ethics Committee. This study was conducted under application number 40713. The results are reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.

CMR Parameters

The UKB Imaging Study began in 2014 and included magnetic resonance imaging of the brain, abdomen, and heart.[17] The CMR protocol consists of a 20-minute scan performed using a 1.5T scanner.[18] We applied a fully convolutional network to CMR scans conducted during the first imaging visit available as of October 8, 2023, to obtain volume and function measures. Exposures of interest were left atrial minimal volume ($LAVI_{min}$)

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and left atrial maximal volume ($LAVI_{max}$) indexed to body surface area, left atrial emptying fraction (LAEF), right atrial minimal volume and right atrial maximal volume indexed to body surface area, and right atrial emptying fraction. The CMR and quality control protocol are described in Methods S1.

Covariates

All covariates were assessed at the time of the imaging visit, with the exception of the Townsend index and biochemical measures.[18–20] The Townsend index was assessed at the baseline visit, and the biochemical measures were taken from the visit closest to the imaging visit. Blood samples were collected from participants during the center examination and processed at a central processing laboratory.[21] HDL (high-density lipoprotein), total cholesterol, and CRP (C-reactive protein) were assayed in all participants.[22] PR interval and P-wave duration measurements were obtained from the imaging visit ECG.

Genetic Data

488 377 participants in the UKB have been genotyped for over 800 000 variants using DNA extracted from blood samples collected at the baseline examination.[23] Genotypes were directly called using the UKB Axiom and UK Applied Biosystems Array. Imputation was performed using the Haplotype Reference Consortium, the UK10K reference panel, and the 1000 Genomes Project reference panel.[23]

Outcomes

Longitudinal follow-up for outcomes was performed through linkages with national databases on hospitalizations, primary care records, and death records. Incident AF events and dementia were identified from self-report and diagnosis codes, while ischemic stroke and heart failure were identified solely from diagnosis codes (Methods S2). Participants diagnosed with AF before the imaging visit and the presence of a pacemaker were excluded from analyses (Figure 1). In addition, participants whose imaging visits occurred after UKB data censoring dates were excluded. The follow-up for this study extended through 2022.

Statistical Analyses

Cox proportional hazard regression models were used to test the association of atrial measures as continuous variables with incident outcomes, excluding individuals with prevalent disease at baseline and adjusting for established risk factors for each outcome. Incident AF models were adjusted for age, sex, ethnicity, body mass index (BMI), systolic blood pressure (SBP), diastolic blood pressure, antihypertensive medication use, current smoking, diabetes, history of heart failure, and history of myocardial infarction. Incident ischemic stroke models were adjusted for age, sex, ethnicity, BMI, SBP, antihypertensive medication use, total cholesterol, HDL, current smoking, and diabetes. Incident heart failure models were adjusted for age, sex, ethnicity, BMI, SBP, antihypertensive medication use, total cholesterol, HDL, left ventricular ejection fraction, diabetes, and history of myocardial infarction. Incident dementia models were adjusted for age, sex, ethnicity, BMI, SBP, antihypertensive medication use, diabetes, Townsend index, and presence of APOE4 (apolipoprotein E4) allele. We used multiple imputation methods with chained equations to account for missing variables.[24] The proportional hazard assumption was evaluated using Schoenfeld residuals within each imputed data set. When the results were inconsistent, we assessed the proportional hazard assumption in a complete case-analysis. In all cases, the global and covariate-specific had $P>0.05$, indicating that the assumption was not meaningfully violated. To explore nonlinear associations, we evaluated each atrial measure as quintiles, adjusting for the same potential confounding variables described above.

Secondary analyses for incident AF, heart failure, and ischemic stroke additionally were adjusted for the Townsend index, CRP levels, and electrocardiographic measures of atrial cardiopathy (P-wave duration and PR interval). Additional secondary adjustment variables for incident dementia included history of ischemic stroke, heart failure, and myocardial infarction. Sensitivity analyses excluded AF that occurred during the first 365 days to exclude potentially undiagnosed AF cases, adjusted for left ventricular ejection fraction, and adjusted for intercurrent AF that occurred after baseline as a time-varying covariate. Effect modification by sex and age was evaluated in fully adjusted models by including multiplicative interaction terms (eg, LAVImin* sex). Separate Cox models were fit for each sex and age subgroup for each outcome. Additional sensitivity analyses were performed with sex-specific standard deviations for models with significant interactions and with adjustment for education as defined by the International Standard Classification of Education categories (range, 1–6) for dementia analyses.[25]

To explore the genetic determinants behind these relationships, we conducted GWAS analyses for each atrial trait, performed genetic correlation analyses, and evaluated causal relationships using several MR approaches (Methods S3). All statistical analysis was performed using R Statistical Software, version 4.0.1 (R Group for Statistical Computing).

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RESULTS

Left and Right Atrial Measures

After excluding participants with prevalent AF, a pacemaker, or lacking follow-up, 51 693 passed CMR quality control and were included in analyses (Figure 1). The mean age was 65, 48% were men, and 24% used hypertension medications (Table S1). The population was relatively healthy and free of cardiovascular disease; only 0.4% had heart failure. For the vast majority of participants, left atrial measures fell within sex-specific normal reference ranges (Table S2). $LAVI_{min}$ was highly correlated with $LAVI_{max}$ and LAEF ($|r|>0.75$), while $LAVI_{max}$ was weakly correlated with LAEF ($r=-0.36$; Figure 1). Similarly, right atrial minimal volume was correlated strongly with right atrial maximal volume and right atrial emptying fraction, while right atrial maximal volume correlated poorly with right atrial emptying fraction ($r=-0.12$). Left-sided measures correlated poorly with right-sided measures ($|r| < 0.3$).

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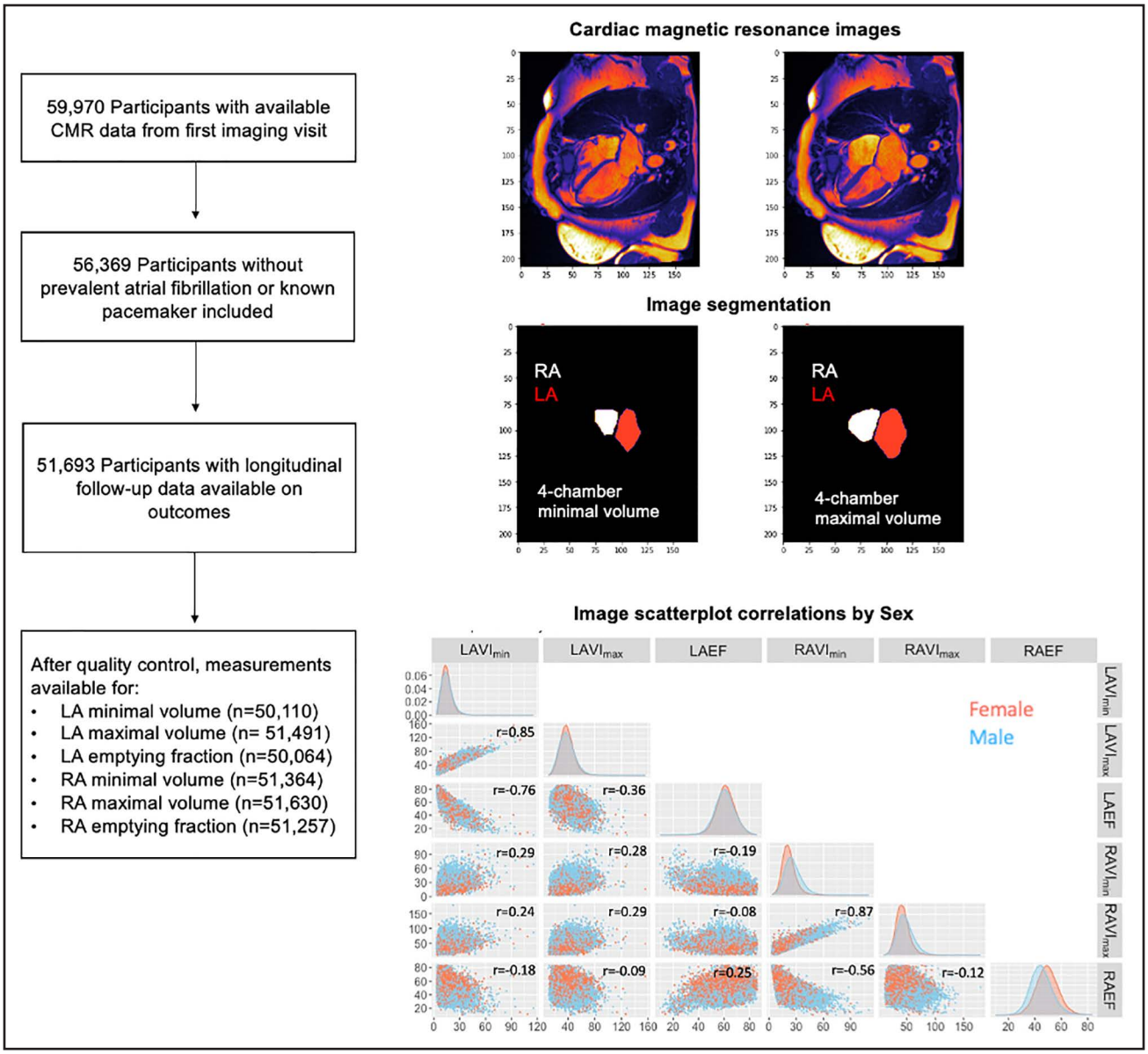

Figure 1. Study flowchart.
After exclusion and quality control, 51 693 participants were included in this study. CMR indicates cardiac magnetic resonance imaging; LA, left atrium; LAEF, left atrial emptying fraction; $LAVI_{max}$, left atrial maximal volume; $LAVI_{min}$, left atrial indexed minimal volume; RA, right atrium; RAEF, right atrial emptying fraction; $RAVI_{max}$, right atrial maximal volume; and $RAVI_{min}$, right atrial minimal volume.

Clinical Associations

During a median follow-up time of 4 (interquartile range, 2.9–5.4) years, 964 (1.9%) participants developed AF, 266 (0.5%) participants developed ischemic stroke, 365 (0.7%) developed heart failure, and 72 (0.1%) developed dementia. In primary analyses adjusting for established clinical risk factors, both left- and right-sided atrial measures were statistically significantly associated with new-onset AF, ischemic stroke, and heart failure (Table 1). Right-sided measures were weaker in magnitude, but several met criteria for statistical significance. For instance, each SD increase in $LAVI_{min}$ was associated with a 55% higher hazard (risk) of AF (95% CI, 48%–62%), while each SD increase in right atrial minimal volume was associated with a 39% higher risk (95% CI, 31%–47%). Only $LAVI_{min}$ was statistically significantly associated with a higher hazard for new-onset dementia (hazard ratio, 1.19 [95% CI, 1.03–1.38]). These associations remained largely unchanged with adjustment for secondary adjustment parameters, left ventricular ejection fraction, and intercurrent AF (Tables S3 through S5).

There was evidence of significant effect modification by sex and age ($P_{interaction}$<0.001; Tables S6 through S9), where several atrial measures were more strongly associated with AF and heart failure in women than in men, many of which remained significant after

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Table 1. Associations of Atrial Function and Structure With Outcomes

Variable	Atrial fibrillation HR (95% CI)	Ischemic stroke HR (95% CI)	Heart failure HR (95% CI)	Dementia HR (95% CI)
$LAVI_{min}$, mL/m²	1.55 (1.48–1.62)*	1.28 (1.16–1.38)*	1.23 (1.15–1.32)*	1.19 (1.03–1.38)*
$LAVI_{max}$, mL/m²	1.60 (1.52–1.68)*	1.19 (1.05–1.35)*	1.35 (1.25–1.46)*	1.21 (0.99–1.48)
LAEF, %	0.55 (0.52–0.59)*	0.75 (0.66–0.85)*	0.76 (0.69–0.84)*	0.85 (0.68–1.05)
$RAVI_{min}$, mL/m²	1.39 (1.31–1.47)*	1.18 (1.02–1.36)*	1.22 (1.10–1.35)*	1.03 (0.85–1.26)
$RAVI_{max}$, mL/m²	1.34 (1.26–1.42)*	1.11 (0.96–1.27)	1.22 (1.09–1.35)*	0.98 (0.80–1.21)
RAEF, %	0.81 (0.74–0.88)*	0.87 (0.76–1.00)	0.93 (0.83–1.03)	0.93 (0.72–1.19)

HRs are reported per SD increase in atrial measure. AF models were adjusted for age, sex, ethnicity, BMI, SBP, DBP, antihypertensive medication use, current smoking, diabetes, history of heart failure, and history of myocardial infarction. Ischemic stroke models were adjusted for age, sex, ethnicity, BMI, SBP, antihypertensive medication use, total cholesterol, HDL, current smoking, and diabetes. Heart failure models were adjusted for age, sex, ethnicity, BMI, SBP, antihypertensive medication use, total cholesterol, HDL, left ventricular ejection fraction, diabetes, and history of myocardial infarction. Dementia models were adjusted for age, sex, ethnicity, BMI, SBP, antihypertensive medication use, diabetes, Townsend index, and APOE4 allele presence. The SD for $LAVI_{min}$ is 6.72, $LAVI_{max}$ is 10.93, LAEF is 8.12, $RAVI_{min}$ is 8.90, $RAVI_{max}$ is 13.48, and RAEF is 9.02. AF indicates atrial fibrillation; APOE4, apolipoprotein E4; BMI, body mass index; DBP, diastolic blood pressure; HDL, high-density lipoprotein; HR, hazard ratio; LAEF, left atrial emptying fraction; $LAVI_{max}$, left atrial maximal volume; $LAVI_{min}$, left atrial minimal volume; RAEF, right atrial emptying fraction; $RAVI_{max}$, right atrial maximal volume; $RAVI_{min}$, right atrial minimal volume; and SBP, systolic blood pressure.

*Significant associations.

correction for multiple hypothesis testing. For example, each SD increase in $LAVI_{min}$ was associated with a 75% higher risk of AF in women compared with a 49% higher risk in men, and a 41% higher risk of heart failure in women compared with a 20% higher risk in men (Tables S6 and S8). The majority of these interactions remained significant when evaluating sex-specific SD increases in each atrial measure (Table S10). When categorizing atrial measures into quintiles, there was a clear threshold effect, where the highest-risk quintile of each left atrial measure was associated with more than a 3× increased risk of AF and a 2× increased risk of heart failure compared with the reference quintile (Figure 2).

Discovery of Independent Genomic Loci for Atrial Traits

Up to 55 881 participants and 11 628 826 variants with minor allele frequency >0.005 were included in the atrial structure and function parameter GWAS. Heritability analyses demonstrated modest genetic contributions to atrial traits (Table S12). We identified 51 significant associations with atrial structure and function traits from 48 variants, of which 27 associations are novel for the specific atrial structure and function traits evaluated (Table 2).[26] Only 2 loci were shared between left and right atrial traits. Ten of the novel loci were located within 500 kb of known AF genes, while 11 have no known link to AF or AF-risk factors (Tables S13 and S14).[27–30] Thirteen of the novel loci were located within 500 kb of known left ventricular genes (Tables S15 and S16). Post-GWAS annotation identified several novel, clinically actionable loci, including rs28456, which was associated with $LAVI_{min}$ and $LAVI_{max}$ and maps to the *FADS1/2* genes, which can be targeted by several omega-3 derivatives that have been extensively studied for their role in reducing adverse cardiovascular events (Table S17).[31–33]

Genetic Correlations

There was a high degree of genetic correlation between chamber-specific minimum and maximal volumes (r=0.94 for $LAVI_{min}$ versus $LAVI_{max}$; r=0.93 for right atrial minimal volume versus right atrial maximal volume; Figure S3). We observed several significant genetic correlations with cardiovascular outcomes and risk factors, with both shared and distinct patterns for left and right atrial traits (Figure S4). Left and right atrial volumes were both positively correlated with increased AF risk. In contrast, chamber-specific patterns were observed for cardiovascular risk factors: left atrial volumes were positively correlated with SBP, indicating shared genetic architecture between left atrial enlargement and high blood pressure, while right atrial volumes were negatively correlated with both SBP and diastolic blood pressure. In addition, right atrial volumes were negatively correlated with type II diabetes, while no significant correlations were observed between left atrial volumes and type II diabetes.

Mendelian Randomization

Using genetic instruments derived from our GWAS findings, forward MR analyses identified causal effects of $LAVI_{max}$ and LAEF on new-onset AF, ischemic stroke, and cardioembolic stroke (Table 3). The strongest association was observed between LAEF and cardioembolic stroke (β=−0.80; P=2.48×10^{-3}). Sensitivity analyses with weighted median estimation demonstrated robustness to assumptions about horizontal pleiotropy (Table S20). There was no evidence of a significant causal association for any of the right atrial measures. In reverse

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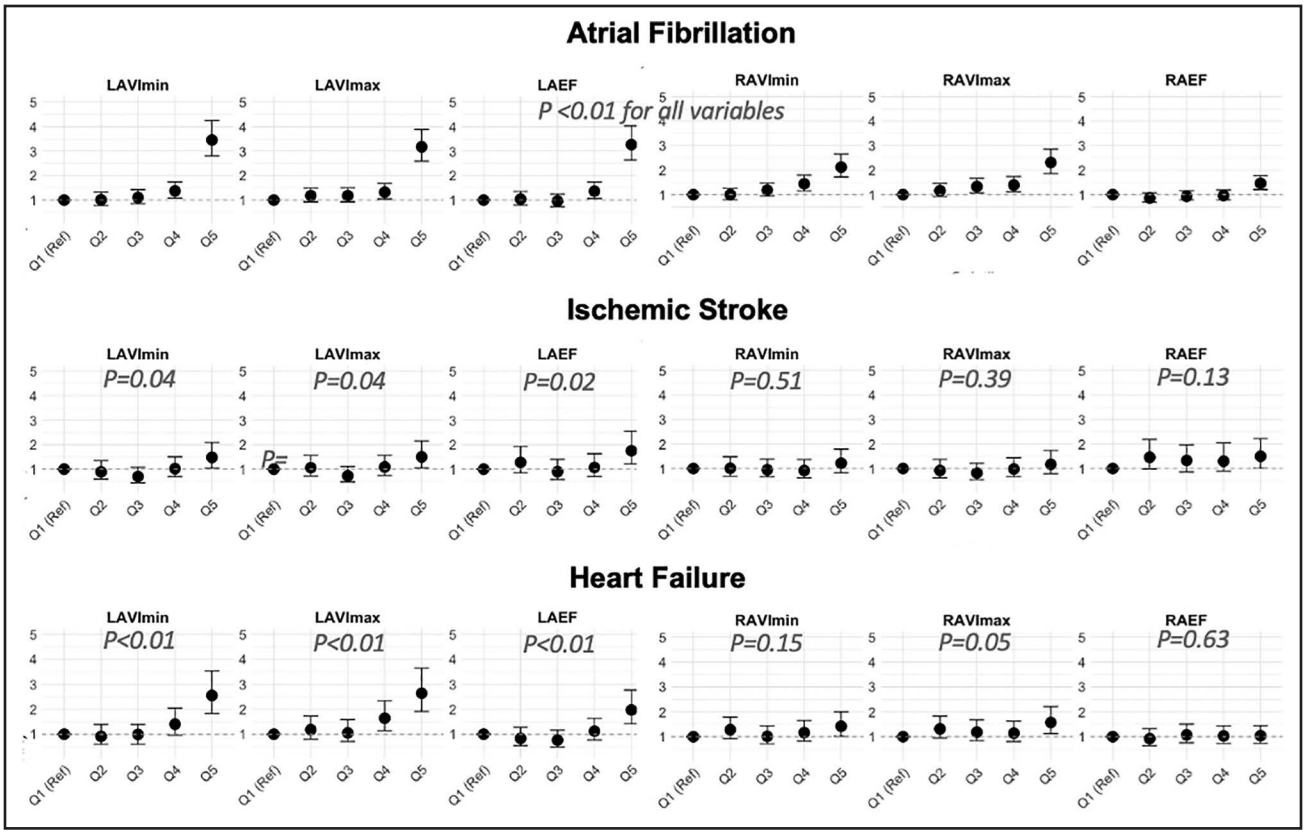

Figure 2. Associations of atrial function and structure with atrial fibrillation, ischemic stroke, and heart failure by quintiles. The highest quintile values of left atrial volume and right atrial volume measures were compared with a reference of the lowest quintile. The lowest quintile values of left atrial emptying fraction and right atrial emptying fraction were compared with a reference of the highest quintile. Hazard ratios are on the *x* axis, and quintiles (Q1–Q5) are on the *y* axis. Atrial fibrillation models were adjusted for age, sex, ethnicity, body mass index (BMI), systolic blood pressure (SBP), diastolic blood pressure, antihypertensive medication use, current smoking, diabetes, history of heart failure, and history of myocardial infarction. Ischemic stroke models were adjusted for age, sex, ethnicity, BMI, SBP, antihypertensive medication use, total cholesterol, HDL (high-density lipoprotein), current smoking, and diabetes. Heart failure models were adjusted for age, sex, ethnicity, BMI, SBP, antihypertensive medication use, total cholesterol, HDL, left ventricular ejection fraction, diabetes, and history of myocardial infarction. *P* values for linear trends by quintile for each variable are reported in gray. LAEF indicates left atrial emptying fraction; $LAVI_{max}$, left atrial maximal volume; $LAVI_{min}$, left atrial indexed minimal volume; RAEF, right atrial emptying fraction; $RAVI_{max}$, right atrial maximal volume; and $RAVI_{min}$, right atrial minimal volume.

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MR analyses, AF and heart failure had causal effects on increased left atrial volumes and decreased LAEF but no effects on right atrial measures. In multivariable MR analyses accounting for the effect of AF, there was no significant evidence of a direct effect of left or right atrial measures on stroke (ie, independent of AF or the genetic mechanisms that lead to AF; Table S21).

DISCUSSION

In this largest cardiac imaging investigation of biatrial cardiopathy to date, both left and right atrial structure and function measures from CMR were associated with new-onset AF, heart failure, and ischemic stroke after adjustment for established risk factors. Despite a small number of events, $LAVI_{min}$ was also associated with new-onset dementia. These associations were robust to adjustment for additional potential risk factors, and many of the associations were substantially stronger in women. Our GWAS of these traits identified 27 novel loci, several of which do not overlap with prior AF loci. These results were used as genetic instruments in MR analyses, which established a strong causal relationship of left atrial measures with AF, ischemic stroke, and cardioembolic stroke, the hallmark complication of AF and atrial cardiopathy, while revealing that right atrial measures are not causally associated with outcomes.

Prior research has established the association between atrial cardiopathy and increased risks of AF, ischemic stroke, and heart failure.[34–37] In the Multi-Ethnic Study of Atherosclerosis, higher $LAVI_{max}$, lower LAEF, and lower peak left atrial strain were independently associated with an increased risk of AF.[8] Similar associations have been reported between reduced LAEF and the risk of ischemic stroke.[7] Previous releases of the UKB imaging data have also been analyzed, with smaller sample sizes, shorter follow-up durations, and a fraction of the cardiovascular events reported here.[19,28]

Table 2. Genetic Loci for Left and Right Atrial Measures

Trait	RSID	Chr	Position	Coded allele	Other allele	CAF	β	SE	*P*value	Gene	Novel	AF gene
$LAVI_{min}$	rs147436345	1	51664488	T	TTTTATTTATTTATTTATTTTA	0.99	−0.19	0.034	3.60×10^{-8}	*LINC01562*	X	X
$LAVI_{min}$	rs9685833	4	111682944	G	A	0.20	0.05	0.008	2.44×10^{-9}	*LINC01438*		X
$LAVI_{min}$	rs35216833	8	97223162	C	T	0.46	0.04	0.006	9.76×10^{-9}	*UQCRB*		
$LAVI_{min}$	rs9664170	10	92692047	T	A	0.42	−0.04	0.006	1.88×10^{-9}	*RNU6-740P*		
$LAVI_{min}$	rs28456	11	61589481	A	G	0.69	0.04	0.007	8.64×10^{-11}	*FADS1*	X	
$LAVI_{min}$	rs2585035	15	65496475	T	G	0.37	0.04	0.006	1.45×10^{-8}	*CILP*	X	
$LAVI_{min}$	rs3810333	19	46318691	T	C	0.61	0.04	0.007	3.04×10^{-8}	*SYMPK*		
$LAVI_{min}$	rs133902	22	26164079	C	T	0.56	−0.04	0.006	2.79×10^{-11}	*MYO18B*		X
$LAVI_{max}$	rs72858515	1	10704547	A	T	0.96	0.10	0.016	1.17×10^{-10}	*CASZ1*	X	X
$LAVI_{max}$	4:111697815_TG_T	4	111697815	TG	T	0.80	−0.04	0.008	3.14×10^{-8}	*LINC01438*		X
$LAVI_{max}$	rs12656497	5	32831939	T	C	0.40	−0.03	0.006	4.19×10^{-8}	*NPR3*		
$LAVI_{max}$	rs12190287	6	134214525	C	G	0.63	−0.04	0.006	6.15×10^{-10}	*TCF21*	X	
$LAVI_{max}$	rs17681427	8	8462368	A	G	0.83	0.05	0.008	3.42×10^{-8}	*RN7SL178P*	X	
$LAVI_{max}$	rs34389718	8	9065525	A	G	0.61	0.04	0.006	7.11×10^{-9}	*AC022784.1*	X	
$LAVI_{max}$	rs3808604	8	10698227	A	T	0.69	0.04	0.007	6.77×10^{-9}	*AC011008.1*	X	
$LAVI_{max}$	rs2645457	8	11614112	T	G	0.42	0.04	0.006	1.85×10^{-9}	*GATA4*	X	X
$LAVI_{max}$	rs28456	11	61589481	A	G	0.69	0.04	0.007	3.16×10^{-11}	*FADS1*	X	
$LAVI_{max}$	rs78341827	15	73401141	C	G	0.96	−0.09	0.016	3.12×10^{-8}	*NEQ1*	X	X
$LAVI_{max}$	rs133873	22	26156505	T	A	0.77	−0.04	0.007	4.08×10^{-8}	*MYO18B*		X
LAEF	rs10801991	1	116309438	T	G	0.66	−0.05	0.006	8.18×10^{-13}	*CASQ2*		X
LAEF	rs35487382	9	97585788	A	G	0.58	0.03	0.006	4.65×10^{-8}	*AOPEP*		X
LAEF	rs10748555	10	92682060	C	A	0.49	−0.03	0.006	1.51×10^{-8}	*RNU6-740P*		
LAEF	rs2681038	15	65497849	A	G	0.37	−0.04	0.006	4.27×10^{-10}	*CILP*		
LAEF	rs10421891	19	46315809	A	G	0.64	0.04	0.006	9.64×10^{-11}	*RSPH6A*		
LAEF	rs133885	22	26159289	G	A	0.55	0.05	0.006	1.73×10^{-15}	*MYO18B*		X
$RAVI_{min}$	rs12465459	2	179672711	T	C	0.34	−0.05	0.006	5.75×10^{-13}	*TTN*		X
$RAVI_{min}$	rs6441111	3	156821808	C	T	0.52	−0.03	0.006	9.76×10^{-9}	*LINC00880*	X	
$RAVI_{min}$	rs4868243	5	172643118	G	A	0.84	−0.05	0.008	4.65×10^{-11}	*RPL7AP33*		X
$RAVI_{min}$	rs12210733	6	118653075	G	A	0.95	0.10	0.013	9.09×10^{-13}	*SLC35F1*		X
$RAVI_{min}$	rs798560	7	2758309	A	G	0.71	0.04	0.007	2.45×10^{-8}	*AMZ1*	X	
$RAVI_{min}$	rs12542381	8	10427495	C	G	0.77	0.04	0.007	9.14×10^{-9}	*PRSS55*	X	
$RAVI_{min}$	rs112852637	8	32413240	T	C	0.53	−0.04	0.006	1.11×10^{-12}	*NRG1*		
$RAVI_{min}$	rs11786896	8	145018354	C	T	0.95	−0.09	0.014	4.65×10^{-10}	*PLEC*	X	
$RAVI_{min}$	rs34005679	10	112458010	T	TC	0.45	−0.04	0.006	2.23×10^{-9}	*RBM20*	X	X
$RAVI_{min}$	rs138893640	12	101498230	C	T	0.99	0.19	0.035	4.75×10^{-8}	*ANO4*	X	
$RAVI_{min}$	rs4766734	12	115155121	T	A	0.65	0.04	0.006	6.33×10^{-12}	*AC026765.3*		X
$RAVI_{min}$	15:98260531_TC_T	15	98260531	TC	T	0.57	0.04	0.006	3.11×10^{-11}	*LINC00923*	X	
$RAVI_{min}$	rs851058	17	41837719	G	A	0.57	−0.03	0.006	2.29×10^{-8}	*SOST*	X	
$RAVI_{min}$	19:38721268_AAAGAG_A	19	38721268	AAAGAG	A	0.47	0.03	0.006	2.74×10^{-8}	*DPF1*	X	
$RAVI_{min}$	rs56096557	19	46310895	A	C	0.65	−0.04	0.006	1.54×10^{-11}	*RSPH6A*		
$RAVI_{max}$	rs12465459	2	179672711	T	C	0.34	−0.04	0.006	2.49×10^{-9}	*TTN*		X
$RAVI_{max}$	rs2277923	5	172662024	T	C	0.70	−0.04	0.007	2.31×10^{-9}	*NKX2-5*	X	X
$RAVI_{max}$	rs3951016	6	118559658	T	A	0.53	0.04	0.006	5.12×10^{-10}	*SLC35F1*	X	X
$RAVI_{max}$	rs150386926	8	32412419	A	AGACT	0.56	−0.04	0.006	1.62×10^{-9}	*NRG1*		
$RAVI_{max}$	rs138893640	12	101498230	C	T	0.99	0.19	0.035	2.11×10^{-8}	*ANO4*	X	

(*Continued*)

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Table 2. Continued

Trait	RSID	Chr	Position	Coded allele	Other allele	CAF	β	SE	*P*value	Gene	Novel	AF gene
$RAVI_{max}$	12:115162091_GTGTGCCCC_G	12	115162091	GTGTGCCCC	G	0.62	0.04	0.006	1.10×10^{-8}	*AC026765.3*	*X*	*X*
$RAVI_{max}$	rs17631783	17	61687600	C	T	0.74	−0.04	0.007	8.57×10^{-10}	*AC046185.2*	*X*	
RAEF	rs146170154	6	36646768	C	CTA	0.80	−0.04	0.008	4.77×10^{-9}	*CDKN1A*	*X*	*X*
RAEF	rs770537744	10	112573688	CAG	C	0.89	0.06	0.010	3.70×10^{-9}	*RBM20*	*X*	*X*
RAEF	rs412768	14	23866713	A	G	0.69	0.05	0.006	4.25×10^{-15}	*MYH6*		*X*
RAEF	rs2854300	19	46274392	G	C	0.96	−0.08	0.015	8.42×10^{-9}	*DMPK*		

Variants with $P<5\times10^{-8}$ from the right and left atrial GWAS are shown. Index variants are identified for each locus by iteratively taking the variant with the lowest *P* value and excluding other variants within a 500-kb window. The nearest genes were annotated using GENCODE, version 26. Novel genetic associations (not within 500 kb of previously reported studies for the individual trait) identified with left and right atrial structure and function are indicated in the Novel column. Genetic associations with atrial fibrillation identified from prior GWAS within 500 kb of our results are indicated in the AF Gene column. AF indicates atrial fibrillation; CAF, coded allele frequency in the UK Biobank cohort; chr, chromosome; GWAS, genome-wide association study; LAEF, left atrial emptying fraction; $LAVI_{max}$, left atrial maximal volume; $LAVI_{min}$, left atrial minimal volume; position, hg19 position; RAEF, right atrial emptying fraction; $RAVI_{max}$, right atrial maximal volume; $RAVI_{min}$, right atrial minimal volume; and β, regression coefficient.

Our results corroborate previous studies that link left atrial volumes with cardiovascular outcomes while reporting several novel significant associations for right-sided measures, which have largely been ignored in prior epidemiological studies, and new evidence of sex-specific findings.

In contrast to previous studies,[38,39] we observed significant sex differences in our findings: for each SD

Table 3. Causal Associations Between Atrial Measures and Outcomes

	$LAVI_{min}$	$LAVI_{max}$	LAEF	$RAVI_{min}$	$RAVI_{max}$	RAEF
Forward direction						
Atrial fibrillation	0.77 (6)	0.72 (8)*	−0.75 (5)	−0.23 (12)	0.77 (6)	0.07 (8)
	P=0.027	$P=6.18\times10^{-3}$*	*P*=0.071	*P*=0.141	*P*=0.026	*P*=0.821
Heart failure	0.05 (6)	0.14 (8)	−0.28 (5)	0.05 (12)	0.05 (6)	−0.05 (8)
	P=0.814	*P*=0.346	*P*=0.094	*P*=0.556	*P*=0.814	*P*=0.822
Ischemic stroke	0.12 (6)	0.2 (8)*	−0.26 (5)*	−0.02 (11)	0.12 (6)	−0.17 (8)
	P=0.0813	$P=2.14\times10^{-3}$*	$P=8.75\times10^{-4}$*	*P*=0.705	*P*=0.081	*P*=0.200
Cardioembolic stroke	0.4 (6)	0.46 (8)	−0.8 (4)*	−0.04 (11)	0.4 (6)	−0.04 (7)
	P=0.291	*P*=0.136	$P=2.48\times10^{-3}$*	*P*=0.737	*P*=0.291	*P*=0.907
Dementia	−0.01 (6)	0.14 (8)	−0.05 (5)	0.02 (12)	0.09 (7)	−0.04 (8)
	P=0.939	*P*=0.257	*P*=0.755	*P*=0.742	*P*=0.426	*P*=0.646
Reverse direction						
Atrial fibrillation	0.07 (138)*	0.06 (146)*	−0.07 (139)*	0.01 (144)	0.02 (146)	0.01 (147)
	$P=3.54\times10^{-9}$*	$P=5.66\times10^{-8}$*	$P=1.86\times10^{-9}$*	*P*=0.198	*P*=0.123	*P*=0.527
Heart failure	0.11 (17)*	0.09 (17)	−0.08 (17)	0.14 (17)	0.12 (17)	−0.08 (17)
	$P=2.43\times10^{-3}$*	*P*=0.033	*P*=0.097	*P*=0.017	*P*=0.047	*P*=0.147
Ischemic stroke	0.05 (46)	0.02 (46)	−0.06 (46)	0.02 (46)	0.01 (46)	−0.05 (46)
	P=0.129	*P*=0.509	*P*=0.068	*P*=0.466	*P*=0.850	*P*=0.084
Cardioembolic stroke	0.03 (13)	0.03 (13)	−0.02 (13)	0 (13)	−0.01 (13)	−0.02 (13)
	P=0.132	*P*=0.078	*P*=0.3871	*P*=0.929	*P*=0.723	*P*=0.518
Dementia	0 (77)	0 (77)	0 (77)	0.02 (77)	0.01 (77)	−0.02 (77)
	P=0.802	*P*=0.871	*P*=0.753	*P*=0.105	*P*=0.191	*P*=0.118

Numbers are β estimates, and the total number of variants included as genetic instruments is in parentheses. For the forward direction MR (atrial measures' effect on outcomes), potential instruments were excluded if the Steiger direction favored systolic blood pressure and diastolic blood pressure with $P<0.05$. No variants met these criteria for body mass index or type II diabetes. For the reverse direction MR of atrial fibrillation's effect on atrial measures, Steiger filtering excluded variants if the Steiger direction favored the atrial measure. For the reverse direction MR of other outcomes' effect on atrial measures, variants with $P<5\times10^{-8}$ for an association with atrial fibrillation were excluded. β coefficients and *P* values were estimated using debiased inverse-variance-weighted methods. LAEF indicates left atrial emptying fraction; $LAVI_{max}$, left atrial maximal volume; $LAVI_{min}$, left atrial minimal volume; MR, Mendelian randomization; RAEF, right atrial emptying fraction; $RAVI_{max}$, right atrial maximal volume; $RAVI_{min}$, right atrial minimal volume; and β, regression coefficient.

*Significant associations with $P<0.010$.

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increase in $LAVI_{min}$, the magnitude of increased AF and heart failure risk was 17% and 18% higher in women than men. These findings align with recent observations that, compared with men, women have more atrial fibrosis and a greater risk of cardiovascular mortality associated with AF and LA enlargement.[40,41] Higher levels of atrial fibrosis in women are observed across nearly all age groups, further suggesting biological differences in atrial remodeling between men and women.[42]

In the largest GWAS of left and right atrial measures to date, we identified 27 novel genetic associations. Prior GWAS of echocardiographic measures of atrial structure failed to discover any significant genetic associations.[43,44] In contrast, previous reports of similarly sized samples from the UKB Imaging Study have identified several loci associated with atrial structure and function, which may be due to higher quality of CMR imaging of the cardiac atria and greater variability in image acquisition with echocardiography.[27–29] In our present GWAS of the largest available sample of UKB CMR data, we discovered 27 novel genetic loci (>500 kb from reported loci) for left and right atrial abnormalities. Some are located near widely replicated AF genes (eg, *PITX2*), while 11 do not map to known AF of AF-risk factor genes.[45]

Linkage disequilibrium score regression identified several genetic correlations with AF-risk factors that differed for left and right atrial traits. Left atrial volumes were positively correlated with SBP, while right atrial volumes were negatively correlated with SBP, diastolic blood pressure, BMI, and type II diabetes. This aligns with observational associations reported in the UKB, where participants with diabetes had lower values of right atrial structure and function in age and sex-adjusted models but no significant difference in left atrial measures.[46] These chamber-specific differences reflect the complex relationship between traditional cardiovascular risk factors and atrial traits, which may be influenced by distinct biological pathways in response to metabolic stress and the unique embryological origins of each atrium. In addition, it is possible that cardiometabolic conditions have direct atrial myocardial involvement, and these differential chamber patterns capture early signs of cardiac dysfunction. Further studies are needed to explore these chamber-specific remodeling mechanisms.

Conducting the new GWAS of atrial traits also improved the strength of our genetic instruments and allowed us to build on previous efforts to estimate the causal effects of left atrial traits on cardiovascular disease risk using MR approaches.[27,28] Disentangling the causal effects of correlated traits (AF, right atrium, and left atrium) that share many, if not most, of the same underlying biological mechanisms can be challenging. In our univariable and multivariable Mendelian analyses, we excluded a large number of genetic variants that impact blood pressure and other known cardiovascular risk factors. Using external genetic evidence to make more realistic or at least explicit assumptions about horizontal pleiotropy may have reduced power and limited the utility of our sensitivity analyses (eg, MR-Egger regression) but overall enhanced the rigor of the causal inference. We did not detect a direct causal effect of abnormalities in atrial traits on ischemic or cardioembolic stroke with currently available genetic evidence, after accounting for the indirect causal effect through AF. However, this fact does not exclude the possibility of clinically meaningful direct causal effects for which our multivariable MR analyses were underpowered.

Strengths of this study include the large sample size, high-quality CMR imaging, and longitudinal follow-up for cardiovascular events identified from UK National Electronic Health Data, which enabled a comprehensive epidemiological and genetic evaluation of atrial cardiopathy. The new GWAS analysis resulted in stronger genetic instruments and improved power for MR analysis, which evaluated the potential causal effects of left and right atrial dysfunction on clinical outcomes. There are also important limitations. First, the vast majority of UKB participants self-identify as White, and our GWAS included only European ancestry individuals to avoid population stratification as a source of bias, which may limit the generalizability of both our epidemiological and genetic study results to other racial/ethnic groups.[47,48] Second, the number of events observed for ischemic stroke and dementia was small, and our median follow-up time of 4 years may have limited our power to identify associations with atrial traits due to small event counts and the natural history of slowly developing conditions such as dementia. Third, the use of self-reported events for AF and dementia cases may underestimate the true prevalence of both conditions, biasing study results.[49,50] However, <5% of cases were determined from self-report. Fourth, our genetic instrument threshold was less stringent ($P<10^{-6}$) than conventional GWAS significant thresholds but allows for exploring potential associations that can guide future validation. Finally, this study uses CMR at a single imaging visit and does not allow us to assess how changes in these measures impact longitudinal outcomes.

CONCLUSIONS

In this analysis of CMR imaging data in over 50 000 participants of a large prospective cohort study, we found that accurate and reproducible measures of both left- and right-sided atrial cardiopathies were associated with an increased risk of AF, ischemic stroke, and heart failure, particularly in women and independent of established clinical risk factors. We also identified several novel genetic loci for these traits and observed chamber-specific patterns of genetic correlation with AF-risk factors that suggest distinct left and right atrial roles

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in the pathogenesis of AF and related cardiovascular complications.

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Supplemental Material

Methods S1–S3
Tables S1–S21
Figures S1–S4
References 51–75

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